Computational Physics I 5630 Fall 2026

Homework Assignment 3

1. One-dimensional random walk with random step length. Consider a one-dimensional random walker whose position after n steps is

$$X_n = \sum_{i=1}^{n} \ell_i, \qquad X_0 = 0,$$

where the steps ℓ_i are independent random variables drawn from a symmetric probability distribution $\pi(\ell)$. Choose **one** of the following two step distributions:

(A) Two-sided exponential: $$\pi(\ell) = \frac{1}{2a} \exp\left(-\frac{|\ell|}{a}\right), \qquad -\infty < \ell < \infty,$$

(B) Semicircle: $$\pi(\ell) = \begin{cases} \dfrac{2}{\pi a^2}\sqrt{a^2 - \ell^2}, & |\ell| \le a, \\ 0, & |\ell| > a. \end{cases}$$

Generate the random steps using an appropriate sampling method discussed in class. Briefly explain the method you use. Using a large ensemble of independent random walkers, carry out the following numerical studies:

1. **Position distribution.** Compute the histogram of the walker position X_n at several different times n. Compare your numerical distributions with the analytical diffusion result obtained in part 4.

2. **Spreading of the random walk.** Compute the standard deviation

   $$\sigma_X(n) = \sqrt{\langle X_n^2 \rangle - \langle X_n \rangle^2}$$

   as a function of n. Plot σ_X versus n and determine its dependence on n from your numerical results.

3. **Dependence on step length.** Choose a fixed value of n and repeat the simulation for several values of the scale parameter a. Plot σ_X versus a and compare your numerical result with an analytical calculation based on your chosen step distribution.

4. **Connection to the diffusion equation.** The probability density $\pi(x,t)$ for one-dimensional diffusion satisfies

   $$\frac{\partial \pi(x,t)}{\partial t} = D \frac{\partial^2 \pi(x,t)}{\partial x^2}.$$

   Determine the diffusion constant D appropriate for your chosen step distribution. Starting from $\pi(x,0) = \delta(x)$, obtain the analytical solution $\pi(x,t)$ and compare it with your numerical histograms in part 1.

Today's in-class checkpoint. Choose one of the two step distributions above. Implement (i) a sampler for the random step ℓ, and (ii) the random-walk simulation. Produce at least one normalized histogram of X_n for a chosen value of n. To obtain sufficient statistics for the histogram, you may either run a one-walker simulation many independent times, or simulate many independent walkers simultaneously (and repeat the calculation a few times if needed).

Checkpoint submission: Please submit today's checkpoint result to **Canvas** by the end of today. Preferably, show me your preliminary result before the end of class.